

Digital Circuits (I)

THE INVERTER

Outline

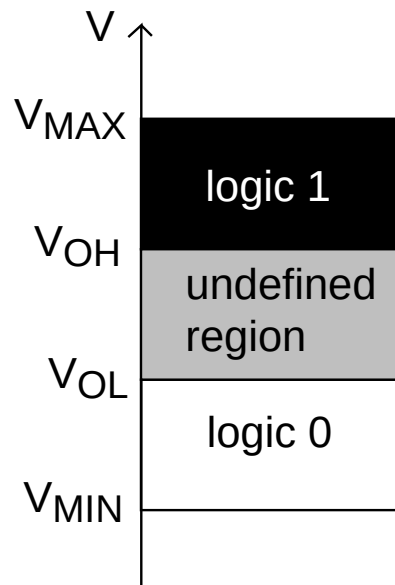
- Introduction to digital circuits
 - The inverter
- NMOS inverter with resistor pull-up

Reading Assignment:

Howe and Sodini; Chapter 5, Sections 5.1-5.3

1. Introduction to digital circuits: the inverter

In digital circuits, digitally-encoded information is represented by means of two distinct voltage ranges:



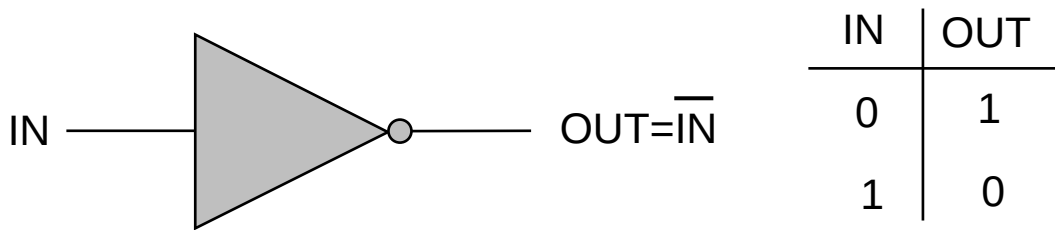
The Static Definition

- **Logic 0:** $V_{MIN} \leq V \leq V_{OL}$
- **Logic 1:** $V_{OH} \leq V \leq V_{MAX}$
- **Undefined logic value:** $V_{OL} \leq V \leq V_{OH}$

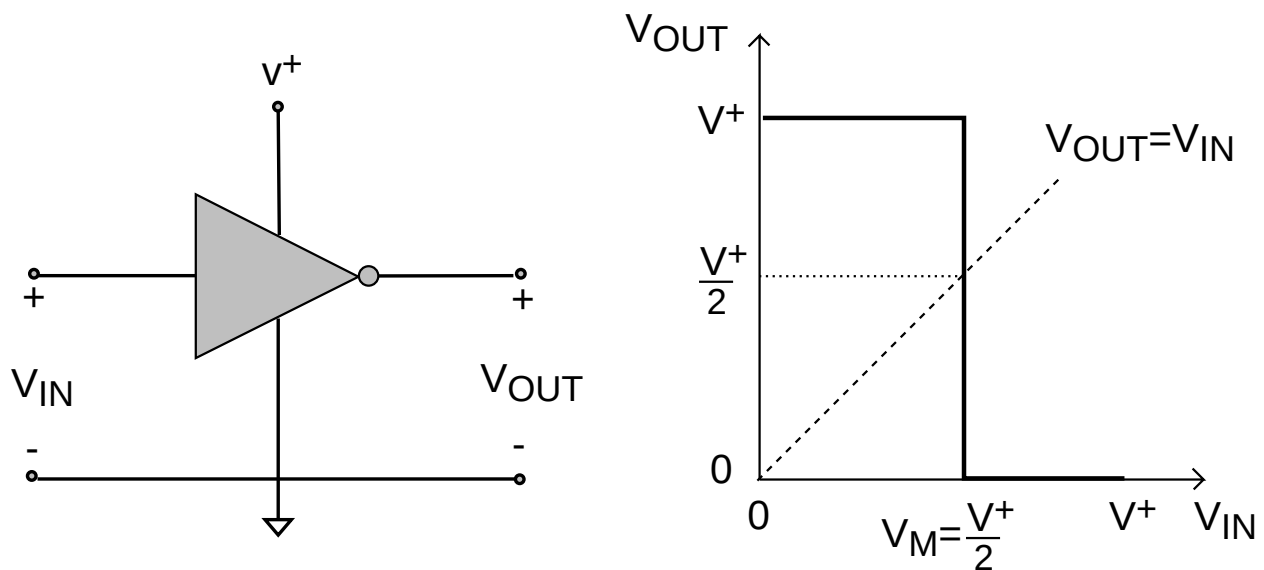
Logic operations are performed using **logic gates**.

Simplest logic operation of all: **inversion** $\Rightarrow$ inverter

Ideal inverter



Circuit representation and ideal transfer function:



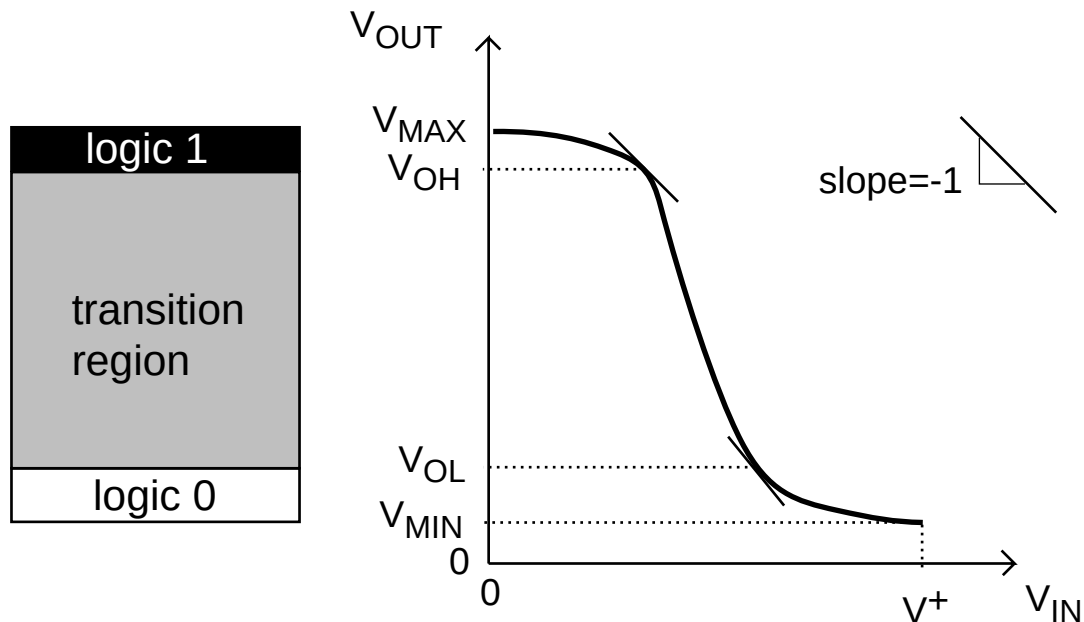
Define **switching point** or **logic threshold** :

- $V_M \equiv$ input voltage for which $V_{OUT} = V_{IN}$
 - For $0 \leq V_{IN} < V_M \Rightarrow V_{OUT} = V^+$
 - For $V_M < V_{IN} \leq V^+ \Rightarrow V_{OUT} = 0$

Ideal inverter returns well defined logical outputs (0 or V^+) even in the presence of considerable noise in V_{IN} (from voltage spikes, crosstalk, etc.)

$\Rightarrow$ signal is **regenerated**!

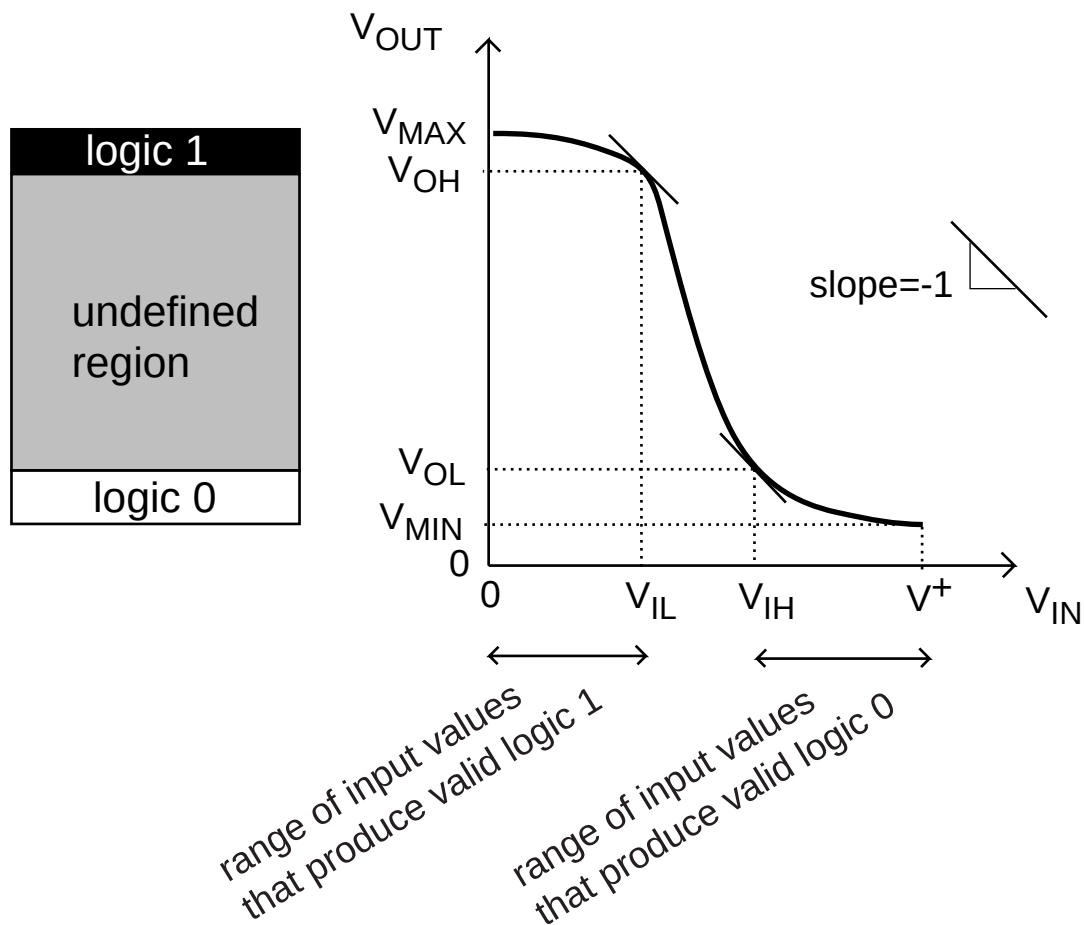
“Real” inverter



In a real inverter, valid logic levels defined as follows:

- Logic 0:
 - $V_{MIN} \equiv$ output voltage for which $V_{IN} = V^+$
 - $V_{OL} \equiv$ smallest output voltage where slope = -1
- Logic 1:
 - $V_{OH} \equiv$ largest output voltage where slope = -1
 - $V_{MAX} \equiv$ output voltage for which $V_{IN} = 0$

Two other important voltages:



Define:

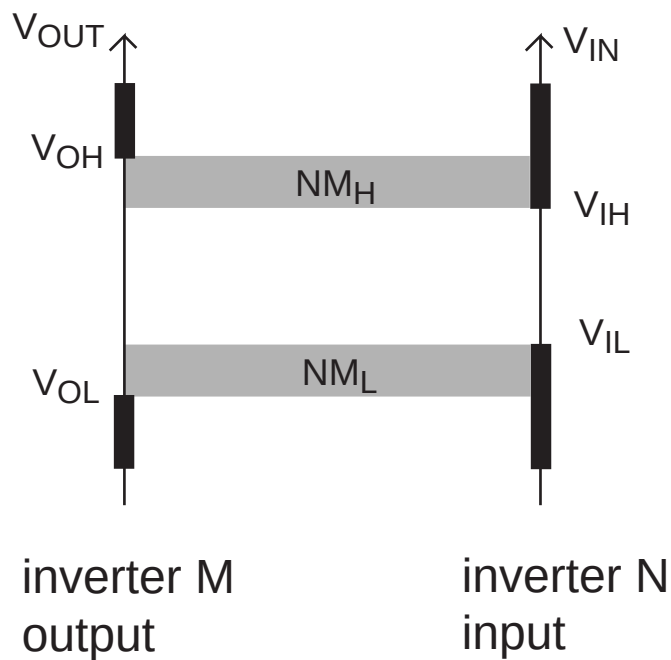
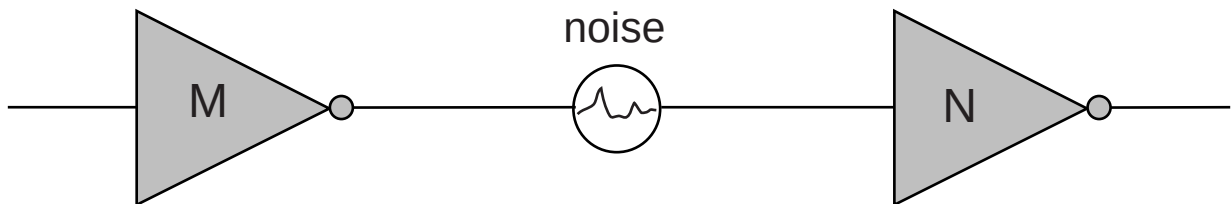
$V_{IL} \equiv$ smallest input voltage where slope = -1

$V_{IH} \equiv$ highest input voltage where slope = -1

If range of output values V_{OL} to V_{OH} is **wider** than the range of input values V_{IL} to V_{IH} , then the inverter exhibits some noise immunity. ($|\text{Voltage gain}| > 1$)

Quantify this through **noise margins**.

Chain of two inverters:



Define **noise margins**:

$$NM_H \equiv V_{OH} - V_{IH}$$

$$NM_L \equiv V_{IL} - V_{OL}$$

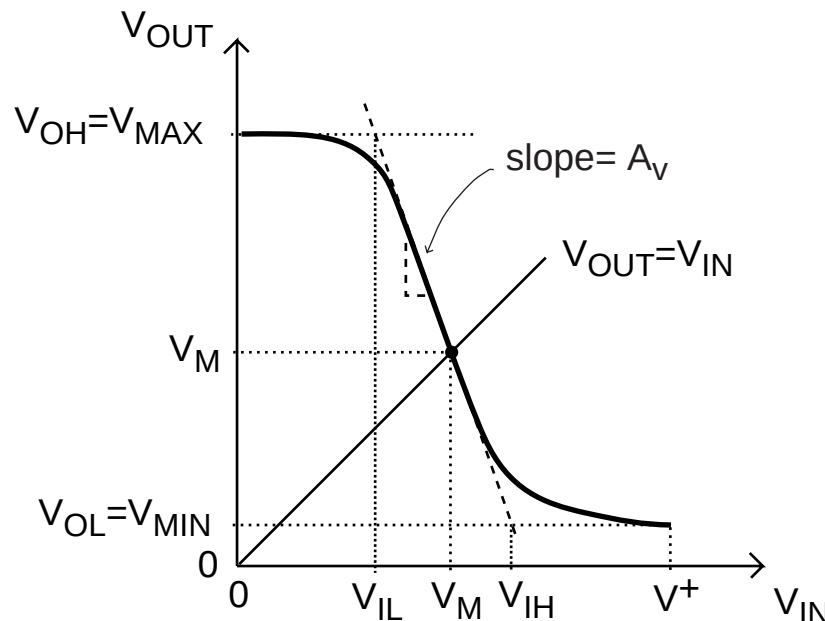
noise margin high

noise margin low

Simplifications for hand calculations: Logic levels and noise margins

It is hard to compute points in transfer function with slope = -1.

Approximate in the following way:



- Assume $V_{OL} \approx V_{MIN}$ and $V_{OH} \approx V_{MAX}$
- Trace tangent of transfer function at V_M
 - Slope = small signal voltage gain (A_v) at V_M
- $V_{IL} \approx$ intersection of tangent with $V_{OUT} = V_{MAX}$
- $V_{IH} \approx$ intersection of tangent with $V_{OUT} = V_{MIN}$

The CMOS Inverter

Slides adapted from:
N. Weste and D. Harris, ***CMOS VLSI Design***
Addison-Wesley, 3/e, 2004

Outline

- Robustness of CMOS Inverter – The Static Behavior
 - Switching threshold
 - Noise Margins
- Performance of CMOS Inverter – Dynamic Behavior
 - Propagation delay
- Power Dissipation
 - Static dissipation
 - Dynamic dissipation

Q&A

1. If the width of a transistor increases, the current will
increase decrease not change
2. If the length of a transistor increases, the current will
increase decrease not change
3. If the supply voltage of a chip increases, the maximum transistor current will
increase decrease not change
4. If the width of a transistor increases, its gate capacitance will
increase decrease not change
5. If the length of a transistor decreases, its gate capacitance will
increase decrease not change
6. If the supply voltage of a chip increases, the gate capacitance of each transistor will
increase decrease not change

Q&A

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CMOS Inverter Static Behavior: DC Analysis

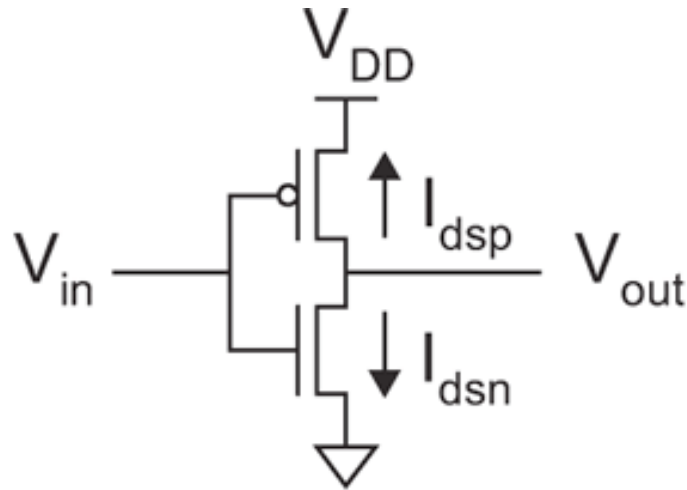


FIG 2.23 A CMOS inverter

CMOS Inverter: DC Analysis

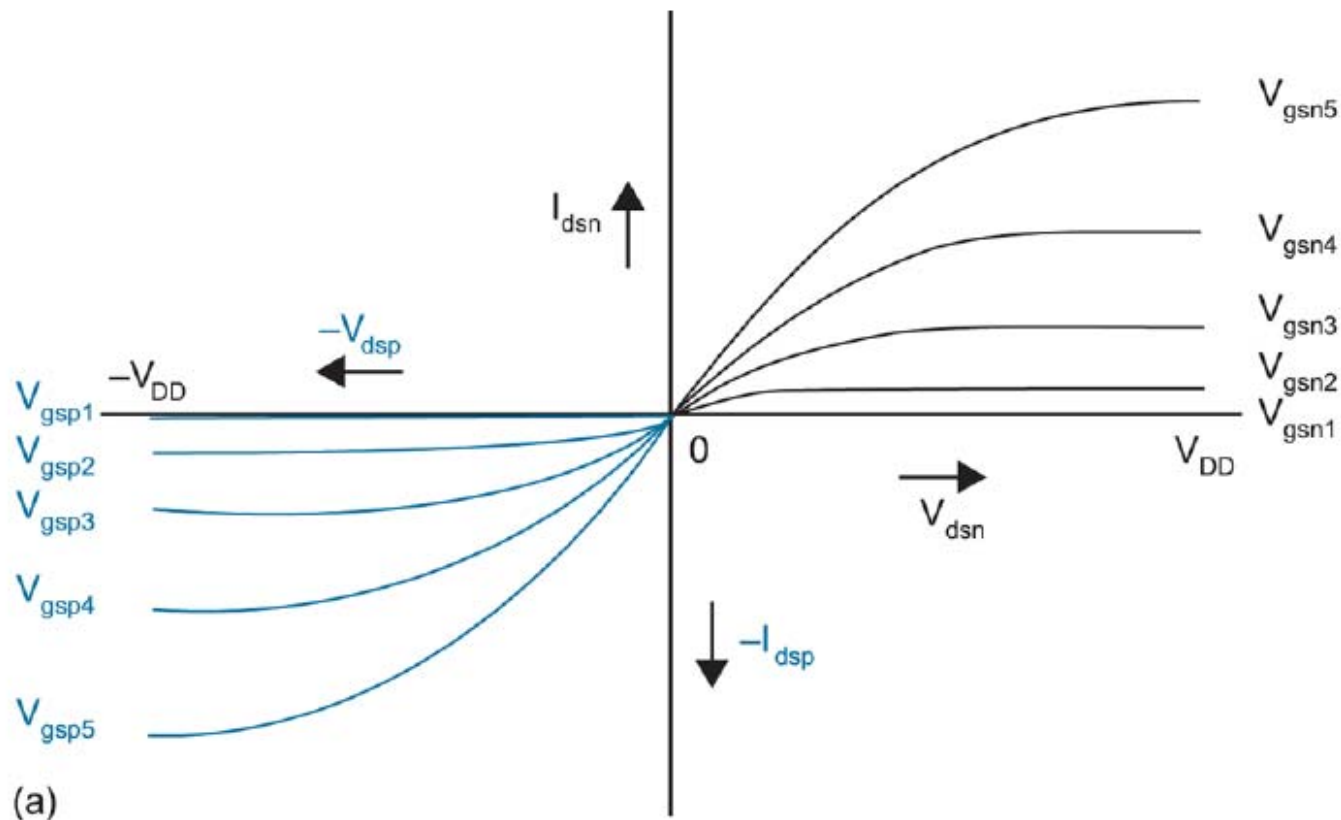
- DC Response: V_{out} vs. V_{in} for a gate
- Inverter
 - When $V_{\text{in}} = 0 \rightarrow V_{\text{out}} = V_{\text{DD}}$
 - When $V_{\text{in}} = V_{\text{DD}} \rightarrow V_{\text{out}} = 0$
 - In between, V_{out} depends on transistor current
 - By KCL, must settle such that
 - $I_{\text{dsn}} = |I_{\text{dsp}}|$
 - We can solve equations
 - Graphical solution gives very good insight

Transistors operation regions

- Current depends on transistor's operation region
- For what V_{in} and V_{out} are nMOS and pMOS in
 - Cutoff ?
 - Linear ?
 - Saturation ?

Graphical derivation of the inverter DC response: I-V Characteristics

- Make pMOS wider than nMOS such that $\beta_n = \beta_p$
- For simplicity let's assume $V_{tn} = -V_{tp}$

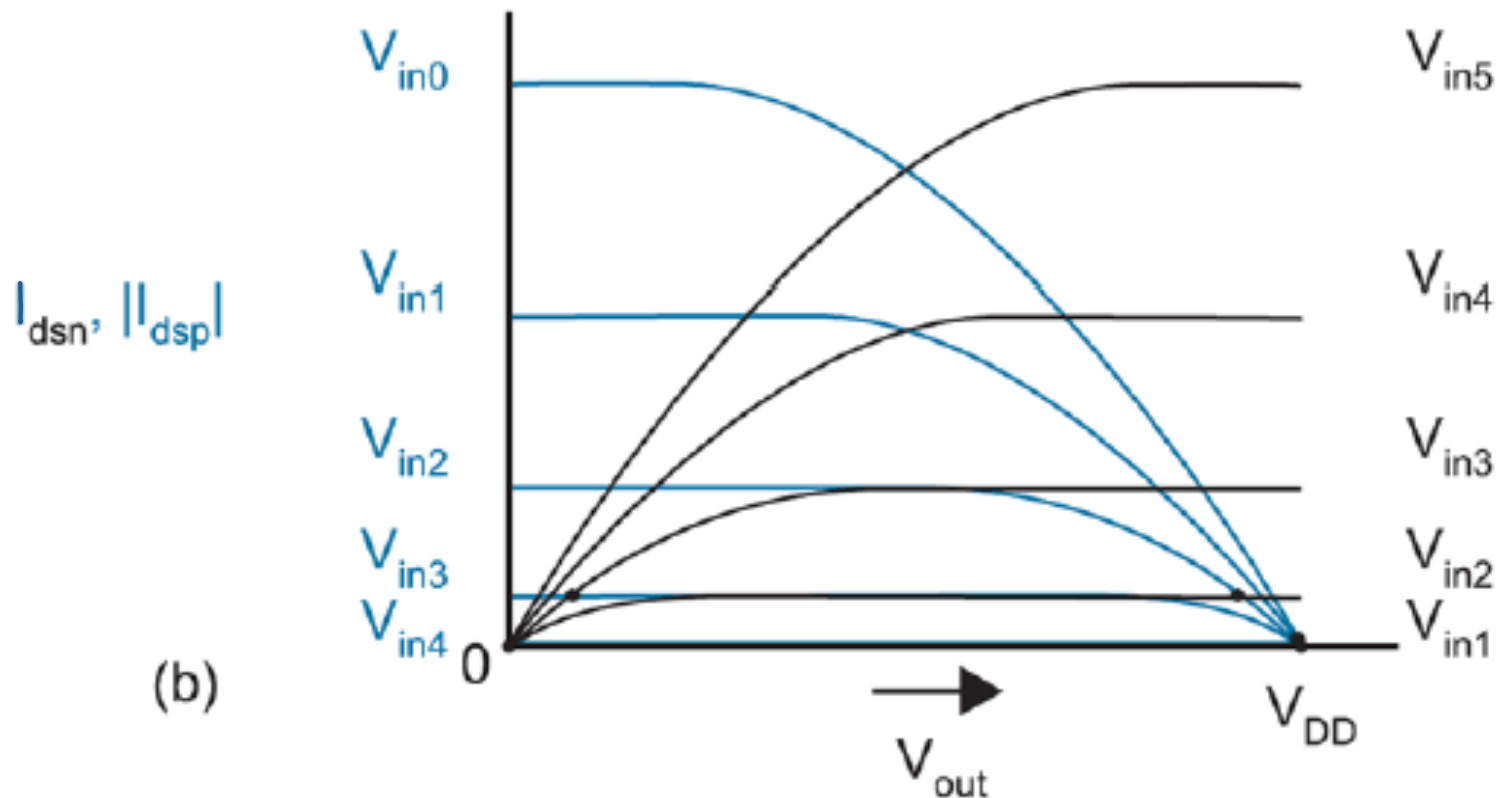


Graphical derivation of the inverter DC response: current vs. V_{out} , V_{in}

- Load Line Analysis:

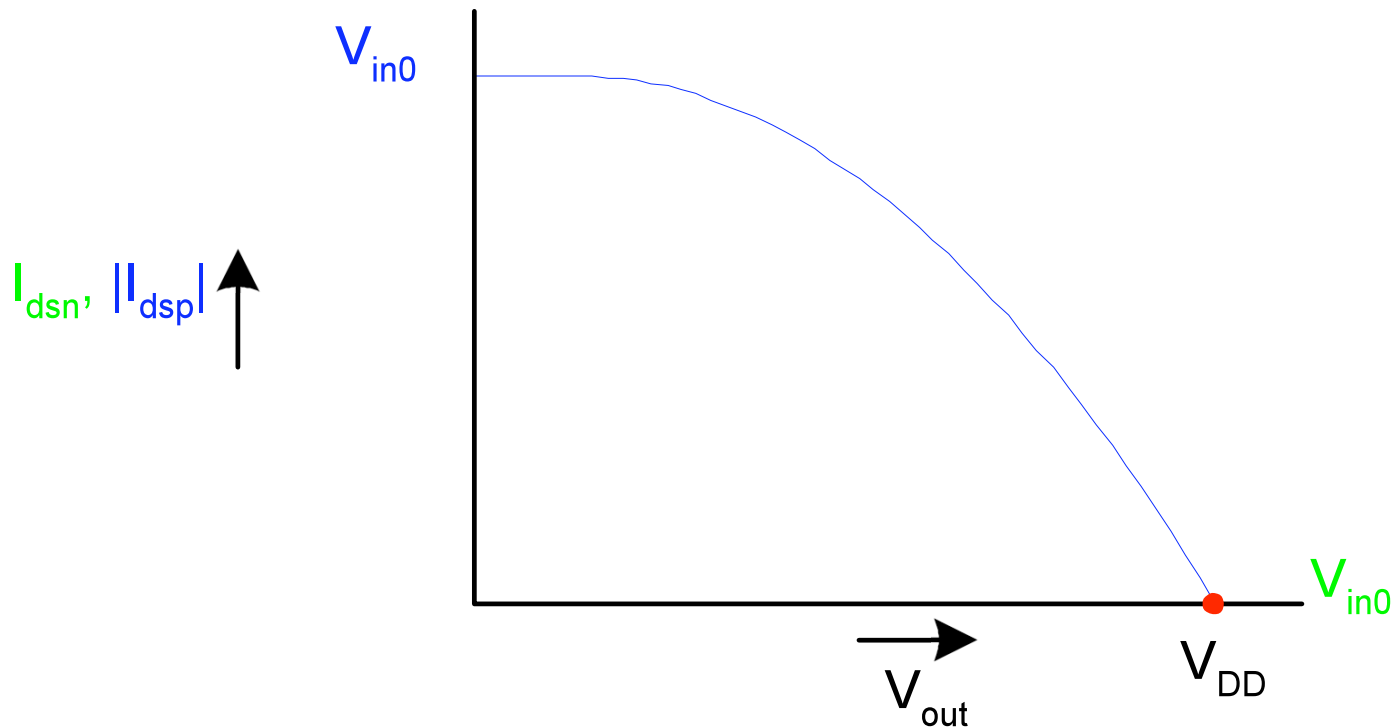
For a given V_{in} :

- Plot I_{dsn} , I_{dsp} vs. V_{out}
- V_{out} must be where $|currents|$ are equal



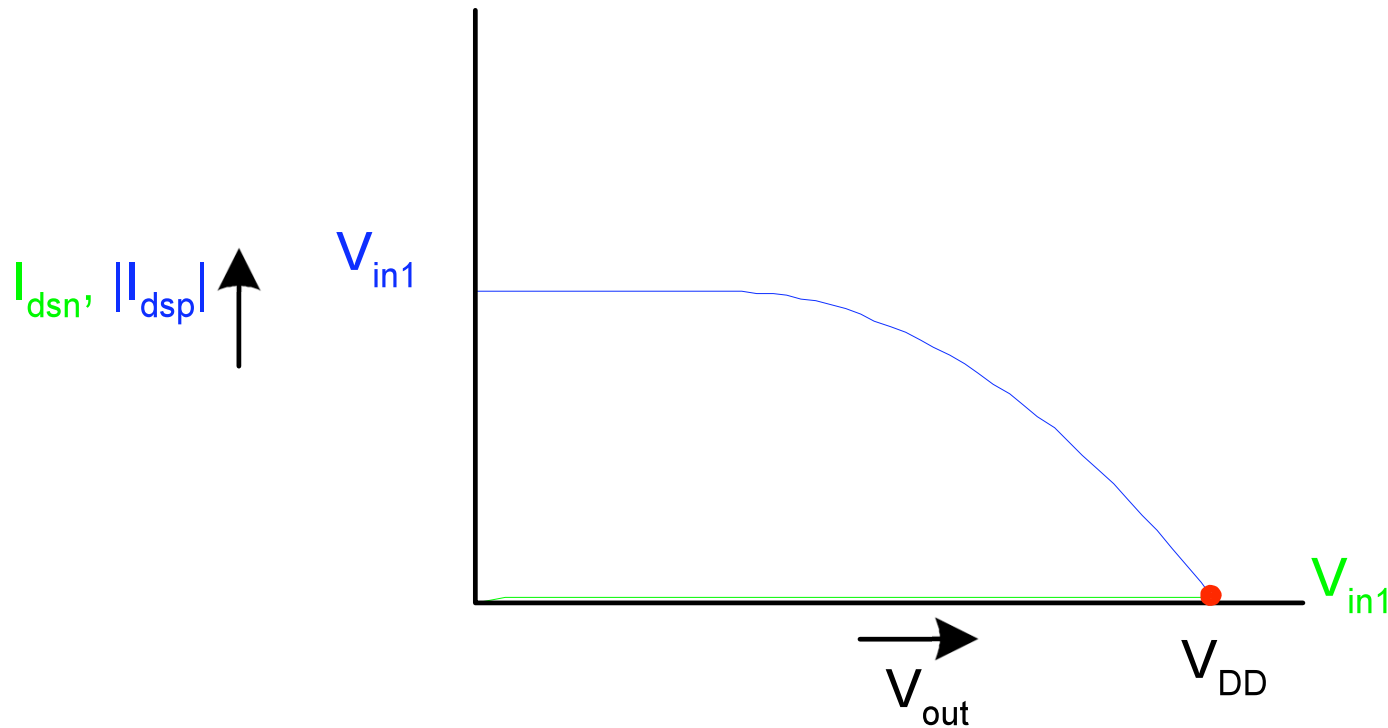
Graphical derivation of the inverter DC response: Load Line Analysis

- $V_{in} = 0$



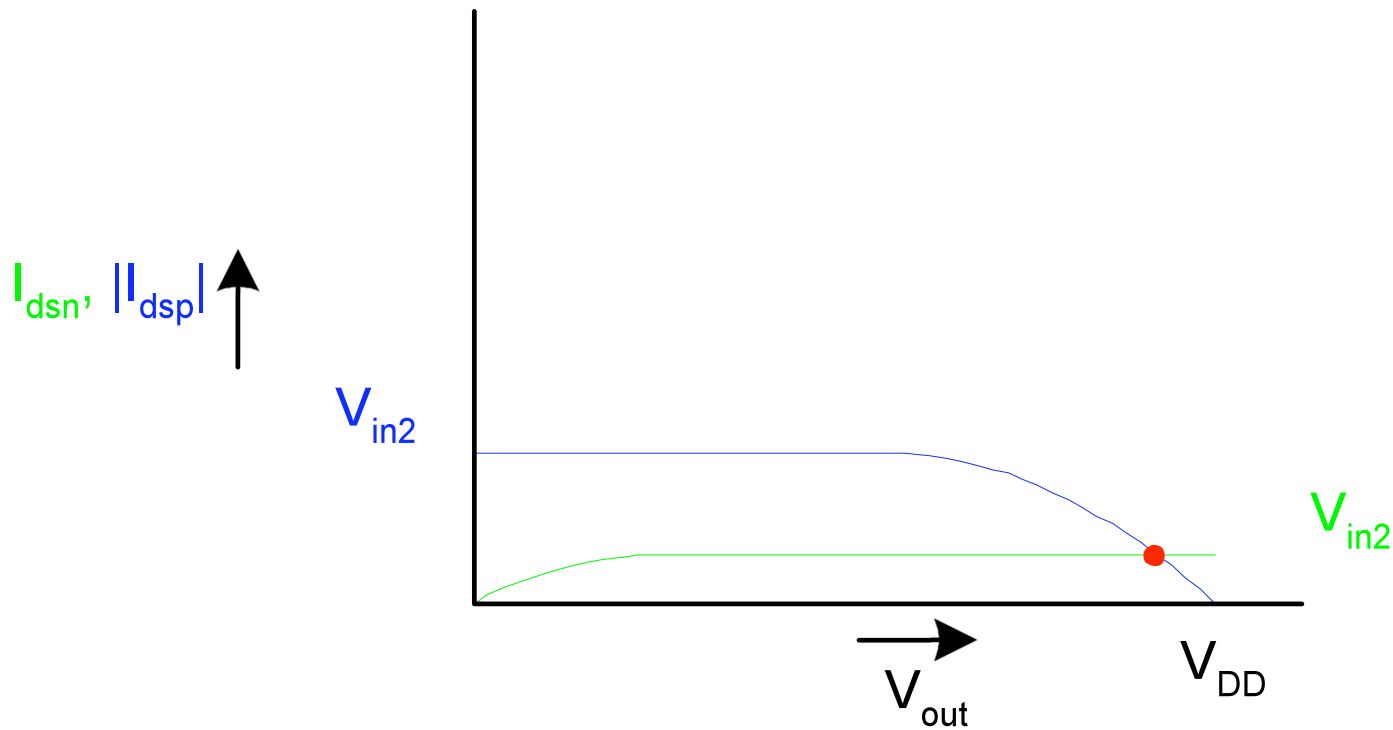
Graphical derivation of the inverter DC response: Load Line Analysis

- $V_{in} = 0.2 V_{DD}$



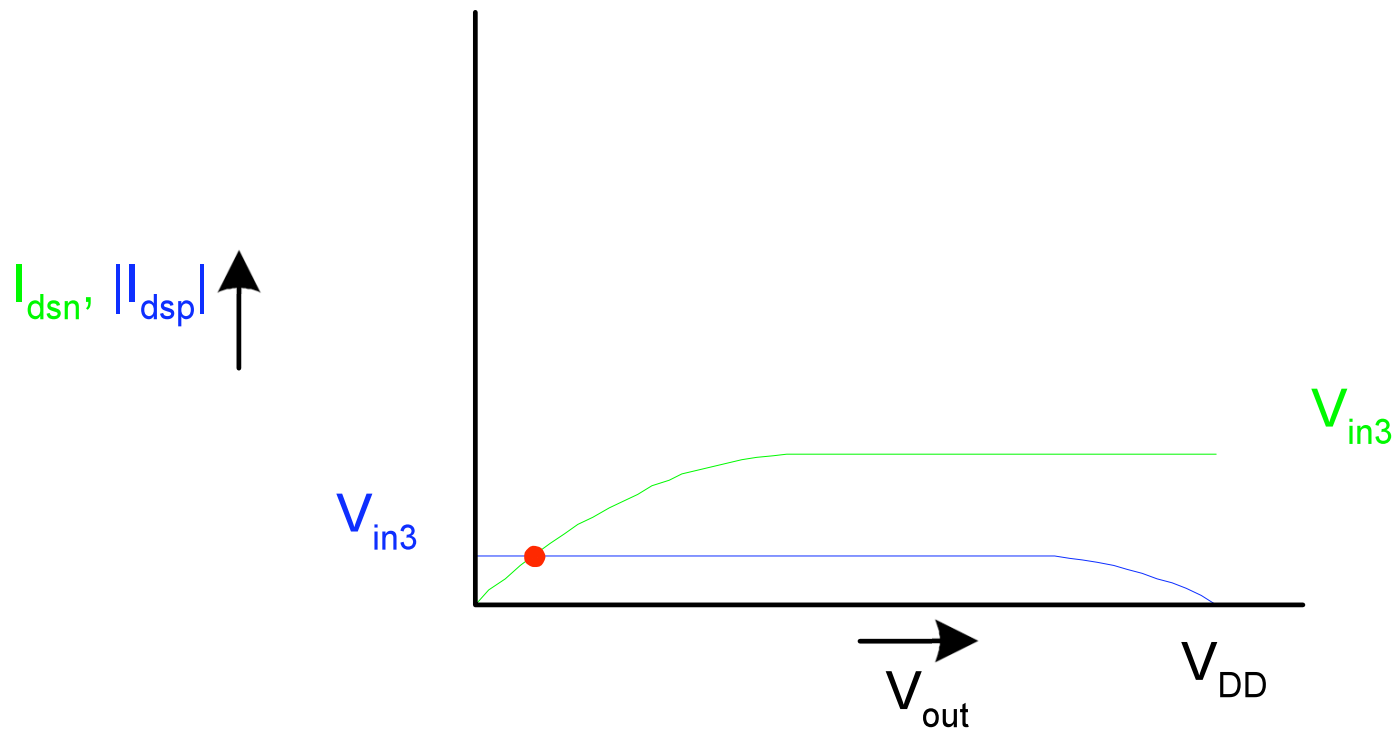
Graphical derivation of the inverter DC response: Load Line Analysis

- $V_{in} = 0.4 V_{DD}$



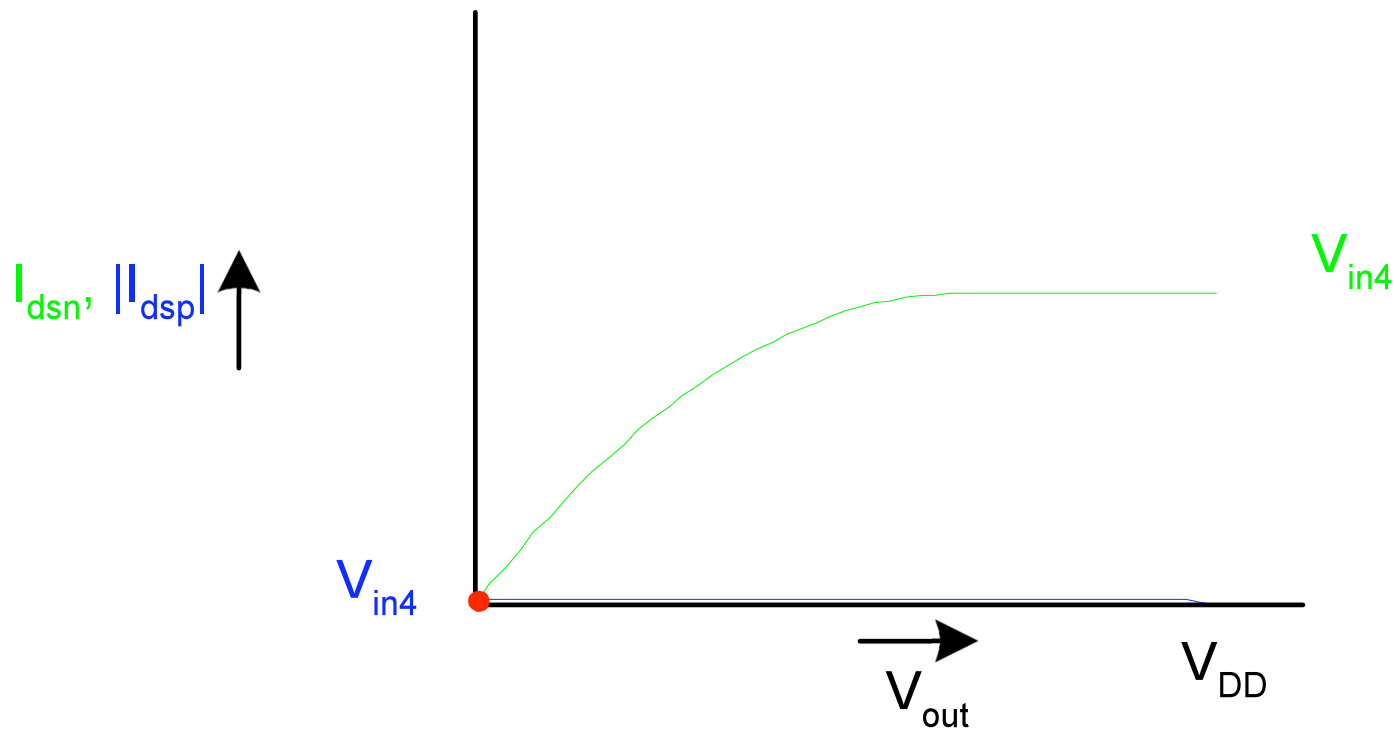
Graphical derivation of the inverter DC response: Load Line Analysis

- $V_{in} = 0.6 V_{DD}$



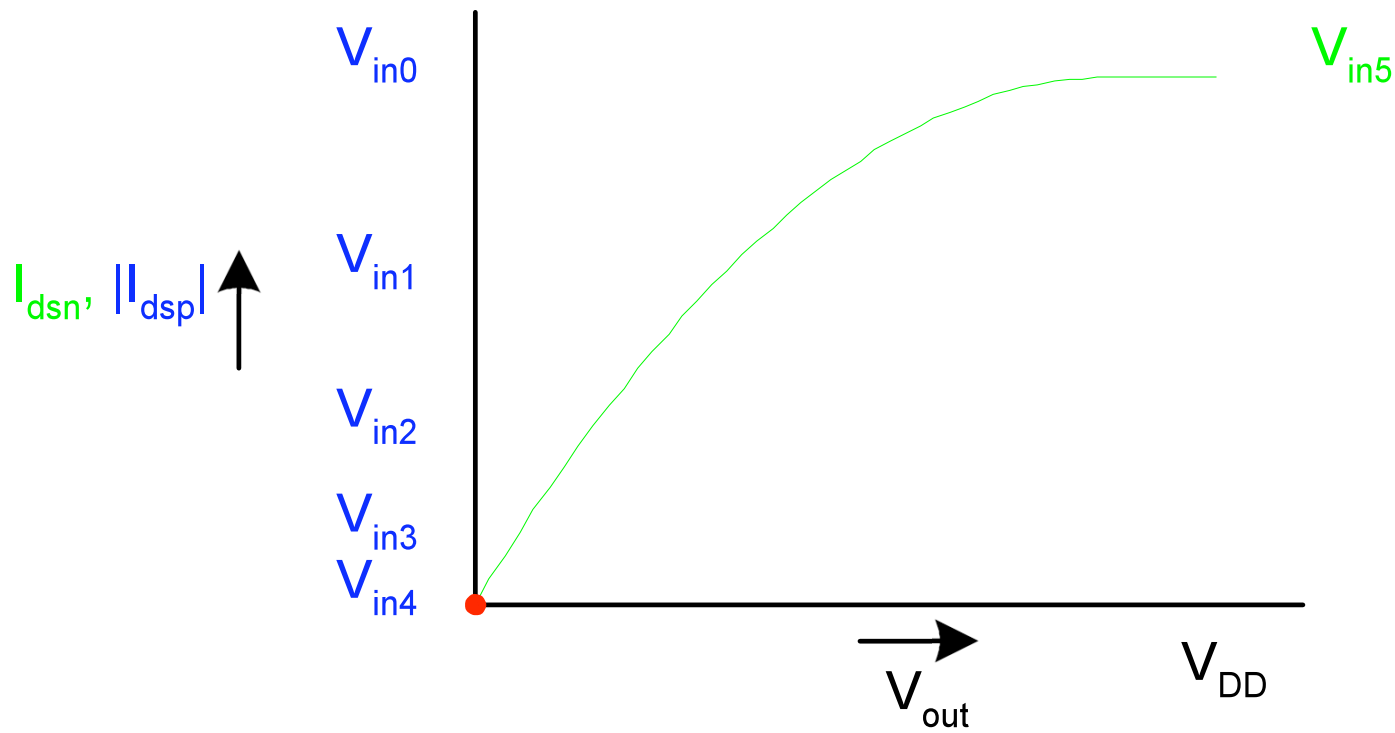
Graphical derivation of the inverter DC response: Load Line Analysis

- $V_{in} = 0.8 V_{DD}$



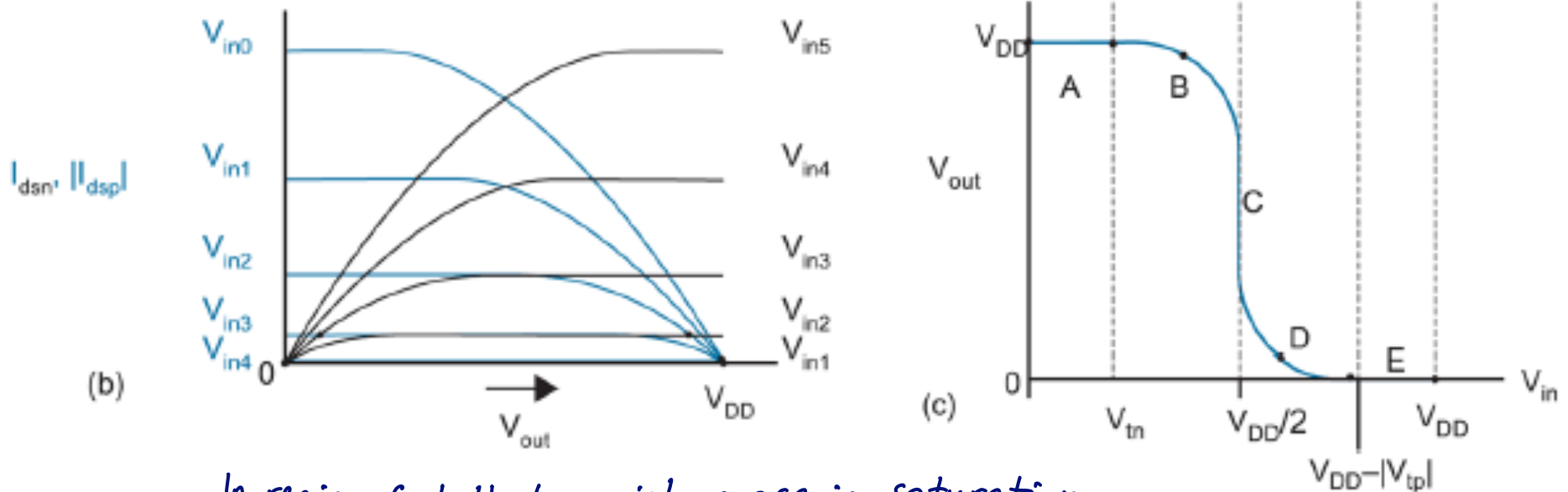
Graphical derivation of the inverter DC response: Load Line Analysis

- $V_{in} = V_{DD}$



DC Transfer Curve

- Transcribe points onto V_{in} vs. V_{out} plot



In region C both transistors are in saturation.
 Ideal transistors are only in region C for $V_{in} = \frac{V_{DD}}{2}$
 and the DC curve slope in C is $-\infty$.

The crossover point where $V_{in} = V_{out}$ is called input threshold.

DC transfer curve: operating regions

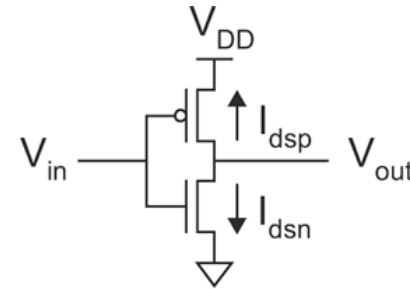
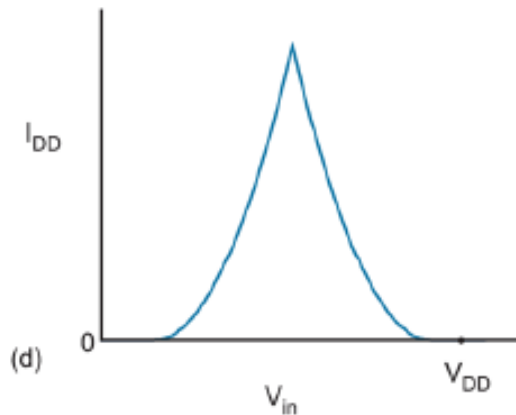
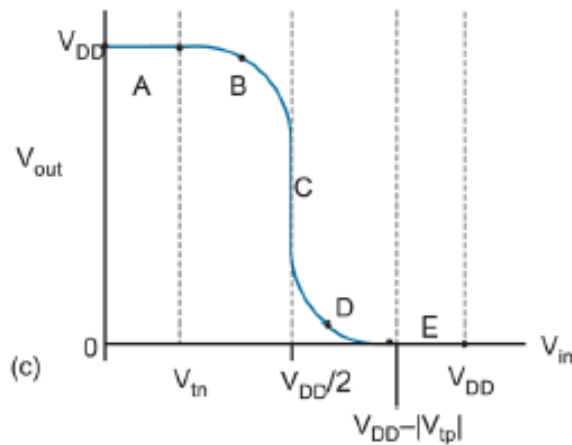


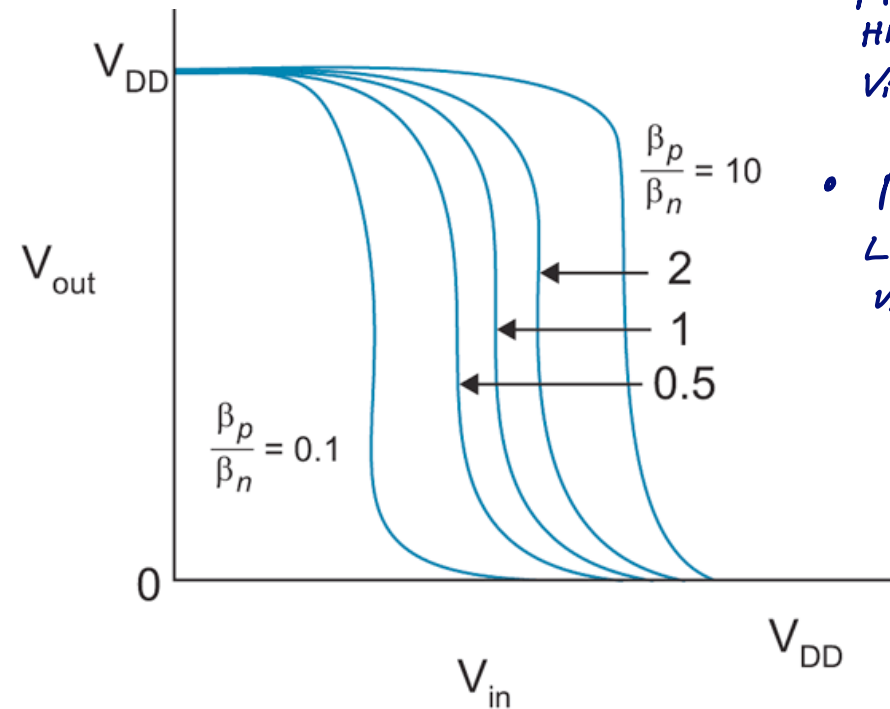
FIG 2.23 A CMOS inverter

Table 2.3 Summary of CMOS inverter operation

Region	Condition	p-device	n-device	Output
A	$0 \leq V_{in} < V_{tn}$	linear	cutoff	$V_{out} = V_{DD}$
B	$V_{tn} \leq V_{in} < V_{DD}/2$	linear	saturated	$V_{out} > V_{DD}/2$
C	$V_{in} = V_{DD}/2$	saturated	saturated	V_{out} drops sharply
D	$V_{DD}/2 < V_{in} \leq V_{DD} - V_{tp} $	saturated	linear	$V_{out} < V_{DD}/2$
E	$V_{in} > V_{DD} - V_{tp} $	cutoff	linear	$V_{out} = 0$

Beta Ratio

- If $\beta_p / \beta_n \neq 1$, switching point will move from $V_{DD}/2$
- Called *skewed gate*



- $\beta_p > \beta_n$
HIGH SKEWED inverter
 $V_{in} = \frac{V_{DD}}{2} \rightarrow V_{out} > \frac{V_{DD}}{2}$
- $\beta_p < \beta_n$
LOW SKEWED inverter
 $V_{in} = \frac{V_{DD}}{2} \rightarrow V_{out} < \frac{V_{DD}}{2}$

FIG 2.26 Transfer characteristics of skewed inverters

Noise Margins

- How much noise can a gate input see before it does not recognize the input ?

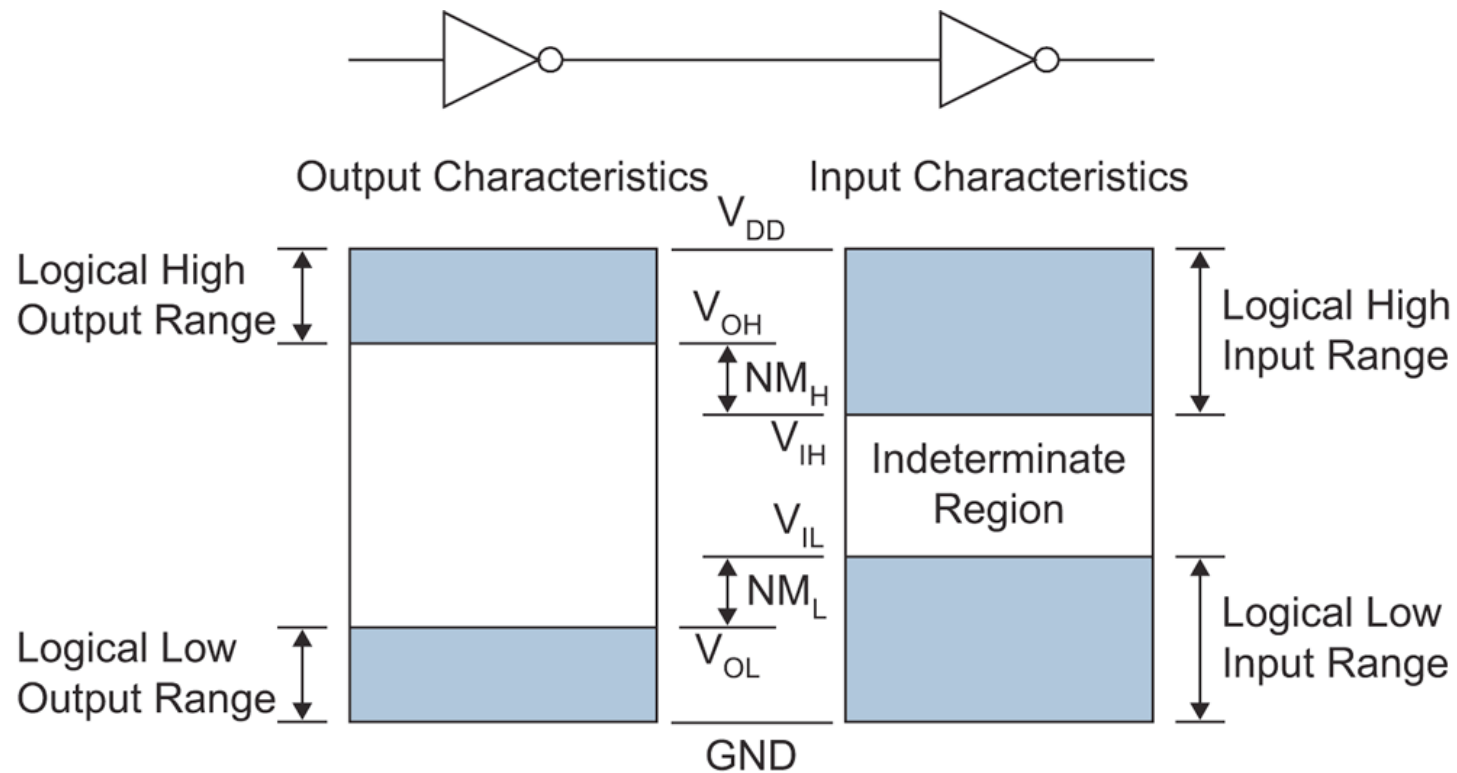


FIG 2.27 Noise margin definitions

Noise Margins

- To maximize noise margins, select logic levels at unity gain point of DC transfer characteristic

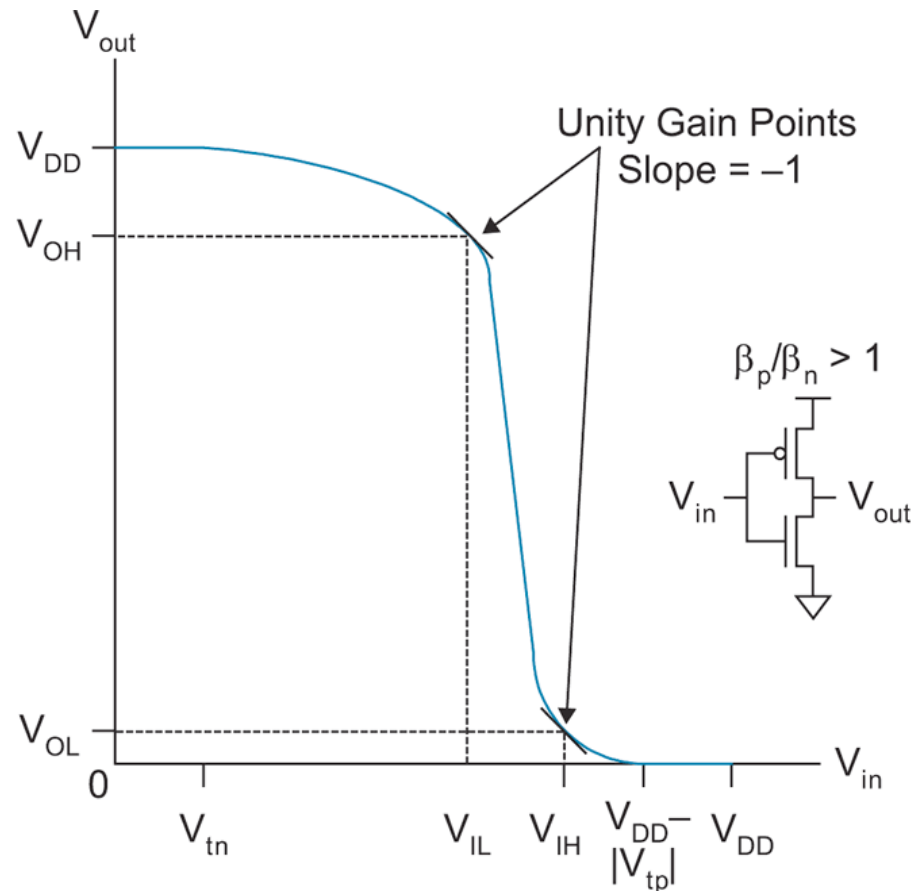


FIG 2.28 CMOS inverter noise margins

DC parameters

- Input switching threshold: V_{TH}
- Minimum high output voltage: V_{OH}
- Maximum low output voltage: V_{OL}
- Minimum HIGH input voltage: V_{IH}
- Maximum LOW input voltage: V_{IL}

} symmetry of the logic gate behavior

} quality of the logic values provided by the gate

} ability of the logic gate to tolerate noise

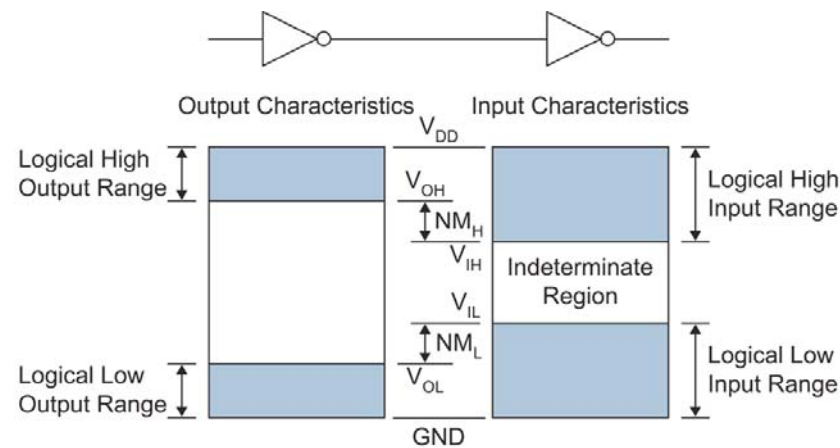


FIG 2.27 Noise margin definitions